Electronic Architecture and Sensor Integration for an Autonomous Multirotor UAV

Abstract—An autonomous multirotor is a tightly integrated electromechanical and computing system. Its flight controller must stabilize the airframe and enforce safety behavior, while positioning sensors estimate vehicle state and higher-level computing interprets the environment and plans tasks. This report reviews the principal electronic components of a quadcopter, including the flight controller, electronic speed controllers (ESCs), brushless motors, propellers, battery and power distribution, global navigation satellite systems (GNSS), radio-control and telemetry links, and a companion computer. It explains the roles and limitations of the inertial measurement unit, magnetometer, barometer, LiDAR, depth camera, and optical flow, and compares thermal imaging and millimeter-wave radar as additional sensing technologies. Common communication interfaces are summarized, and a complete system architecture identifies power, motor-command, sensor-data, RC-control, MAV Link, and image-data paths. Finally, a component-selection example is developed for an outdoor search-and-rescue drone that must navigate waypoints, avoid obstacles, detect people, and report status to a ground station. The central design finding is that reliable autonomy depends less on an individual high-performance component than on compatible interfaces, calibrated sensors, adequate power margins, well-defined fail safes, and end-to-end testing.

I. Introduction

A quadcopter produces lift with four rotors whose speeds are adjusted independently. Differential thrust controls roll, pitch, and yaw; coordinated changes to all rotors control vertical motion. This apparently simple plant requires a fast feedback controller because a multirotor is dynamically coupled and can become unstable if motor commands do not respond to measured motion. Modern autopilot stacks therefore combine sensors, state estimation, control loops, actuator outputs, mission logic, and fault handling in one flight-control subsystem.

Autonomous missions add a second computational problem: the aircraft must interpret position, images, range measurements, and mission goals. A compact flight controller is designed primarily for deterministic real-time control, whereas a companion computer can run heavier workloads such as computer vision, mapping, and path planning. The two cooperate through an explicit interface, commonly MAV Link, rather than replacing one another.

This report follows a system-engineering view. It describes how each component works, what it contributes, where it can fail, and which electrical or software interfaces must be checked during integration. The search-and-rescue example is a conceptual design, not a validated airworthiness assessment. Actual component selection must be confirmed against the aircraft mass, mission duration, environmental conditions, local aviation rules, and the manufacturers' current wiring and safety instructions.

II. System Overview

A. Functional architecture

The system can be divided into five cooperating functions:

- Energy conversion: the battery supplies the DC bus; a power module or power-distribution board routes high current to ESCs and produces regulated rails for electronics.
- State estimation and stabilization: the flight controller fuses IMU, pressure, heading, and positioning information to estimate attitude, altitude, and motion, then computes actuator outputs.
- Propulsion: each ESC converts a low-power command into switched three-phase motor power; the brushless motor and propeller convert electrical energy into thrust.
- Navigation and operator links: GNSS provides global position, the RC link conveys pilot inputs, and telemetry conveys state, mission, and diagnostic information between the aircraft and the ground control station (GCS).
- Perception and mission intelligence: cameras, depth sensors, and/or LiDAR provide environmental data to a companion computer, which can detect people, build maps, and request high-level motion through the flight controller.

B. Control and communication distinctions

Flight controller versus companion computer: the flight controller is the real-time safety-critical control layer. It consums high-rate motion measurements and must still stabilize the vehicle if the companion computer hangs or reboots. The companion computer is an application layer for tasks that need more memory or processing, such as neural-network inference, image processing, simultaneous localization and mapping (SLAM), and route planning. It should send bounded targets or mission requests; the flight controller retains actuator control and the final stabilization loop.

RC versus telemetry: RC communication normally carries pilot stick values and switch states, including arm/disarm and flight-mode inputs. Telemetry carries vehicle position, attitude, battery state, health messages, parameters, and mission commands. They can use separate radios or, in some systems, share a radio link, but they remain distinct functions and need independently reasoned loss behavior. Loss of an

TABLE I
MAJOR DRONE SUBSYSTEMS AND THEIR ROLES

Subsystem	Primary role	Important integration point or limitation
Flight controller	Runs real-time state estimation, stabilization, mode logic, actuator mixing, and failsafe responses.	Firmware support, sensor redundancy, output types, vibration isolation, and verified failsafe configuration matter.
ESCs, motors, propellers	Convert controller output and electrical power into controlled rotor thrust.	Current and voltage ratings, motor/propeller matching, rotor direction, cooling, and thrust margin must be checked together.
Battery and power module/PDB	Supplies the motor bus, regulated avionics power, and voltage/current estimates.	Peak current, voltage compatibility, wiring losses, regulator capacity, and low-battery thresholds affect safe endurance.
GNSS/GPS and compass	Estimate global location, velocity, and heading.	Sky view, multipath, magnetic interference, antenna placement, and correction availability affect performance.
RC receiver and transmitter	Carry pilot commands, arming/mode-switch inputs, and link-quality information.	Receiver protocol, channel mapping, link-loss behavior, and pilot override must be tested.
Telemetry system and GCS	Transfer vehicle status, parameters, mission plans, and commands.	Telemetry loss is not the same as RC loss; link budgets and configured GCS-loss behavior matter.
Companion computer	Runs compute-intensive vision, mapping, planning, and autonomy software.	Power draw, thermal design, boot reliability, interface bandwidth, and safe command bounds must be addressed.
Stabilization sensors	IMU, barometer, magnetometer, GNSS, rangefinder, and optical flow support vehicle-state estimation.	Calibration, timing, alignment, vibration, and sensor health affect estimator quality.
Perception sensors	RGB/thermal cameras, depth cameras, LiDAR, and radar sense people and obstacles.	Field of view, range, lighting/weather sensitivity, latency, mass, and compute requirements affect usefulness.

RC signal may trigger an RC failsafe; loss of the GCS or companion link may trigger a different action. The correct action depends on mission state, GNSS availability, battery reserve, and the firmware configuration.

Manual versus autonomous flight: manual control still commonly relies on the flight controller's stabilization loops; the pilot supplies the desired attitude or rate. In autonomous flight, a mission or companion computer specifies waypoints or high-level setpoints, and onboard control closes the loops to follow them. Modes such as Stabilize, Altitude Hold, Loiter/Position Hold, Auto, Guided, Land, and Return-to-Launch (RTL) represent different levels of automation and different sensor requirements. A manual override and a safe fallback remain important even on a highly autonomous aircraft.

Stabilization versus perception sensing: gyroscopes, accelerometers, barometers, magnetometers, GNSS, and optical-flow/range sensors help estimate the aircraft's state. RGB/thermal imagery, 2D/3D LiDAR, radar, and depth cameras primarily describe the surrounding environment. The boundary is not absolute: range data may help altitude control, and visual odometry may help localization, but perception outputs generally require more processing before they become safe control estimates.

III. ELECTRONIC COMPONENTS

A. Flight controller hardware and firmware

A flight controller (FC), also called an autopilot, is a microcontroller-based board with sensor interfaces, timers, communication ports, actuator outputs, memory, and power regulation. Typical boards incorporate one or more IMUs and a barometer; a magnetometer and GNSS receiver may be integrated or mounted externally. External GNSS/compass placement is common because the antenna needs a clear view of the sky and a compass can be less affected when separated from high-current wires and motors. Pixhawk-compatible hardware is a common ecosystem; examples include Pixhawk 6C/6X-class boards and other manufacturer-supported autopilots. Hardware compatibility should be checked against the chosen firmware release rather than inferred from a product name alone.

Hardware and firmware are different. The board provides the processor, sensors, ports, and outputs. Firmware implements sensor drivers, estimator, controllers, mode logic, actuator allocation, mission logic, logs, and failsafe handling. Ardu Pilot and PX4 are widely used open autopilot software projects. A board may physically resemble another board but still require a specific firmware target, bootloader, pin mapping, or calibration procedure.

In a typical control cycle, the estimator fuses inertial measurements with available heading, altitude, velocity, and position data. Rate and attitude controllers compare estimated state with requested state and calculate corrections. A mixer or control allocator turns total thrust and roll/pitch/yaw demands into individual motor commands. Because control is feedback-based, vibration, bad calibration, wrong motor order, or an incorrect sensor orientation can degrade control even when the software is otherwise functional.

Flight modes vary in their sensor requirements. Stabilize/Acro can operate without global position; altitude-hold modes depend on usable altitude sensing; position hold and waypoint modes generally need a valid position estimate; RTL requires a credible home position and enough energy to return. Flight-mode names and detailed behavior differ between Ardu Pilot and PX4, so the reference guide for the installed release is authoritative.

Failsafe logic may monitor RC signal loss, telemetry/GCS loss, low battery, estimator quality, GPS health, geofence, and other health flags. Typical configured responses include RTL, Land, or switching to a less demanding mode. Failsafe behavior is conditional, not magical: RTL is not a good response when

position is unreliable or the return path is obstructed, and low battery may leave insufficient energy for the intended recovery. Every required trigger and response need a controlled bench or simulation test.

B. Electronic speed controllers

An ESC is the power-electronics stage between the flight controller and a brushless motor. It commutates the motor phases by switching battery current in a controlled sequence. The FC sends a requested output; the ESC regulates phase drive to produce the desired motor speed/torque. Some ESCs also report RPM, voltage, current, temperature, or fault information, but this depends on the ESC and protocol.

Basic legacy ESC control uses pulse-width modulation (PWM), where a pulse width encodes the requested output. Digital protocols such as D Shot encode commands as digital data and can offer improved noise robustness, lower latency, and optional telemetry without traditional analog-style calibration. Drone CAN/CAN-based ESCs communicate over a shared bus and can be attractive for larger or distributed systems. The FC must support the selected protocol on the intended output pins; not every output port supports every protocol.

A separate ESC per motor can make replacement and cooling straightforward, while a 4-in-1 ESC combines four channels in one board, reducing wiring and packaging volume. The latter can simplify a small quad but concentrates heat and creates a shared failure point. ESC selection must account for battery cell count/voltage, continuous and burst current, cooling, connectors, firmware, motor current under the actual propeller load, and connector/wire losses. A reasonable engineering practice is to size current capacity with margin above measured or manufacturer-specified peak operating current, then validate temperatures under a representative load. Ratings for short bursts must not be mistaken for continuous ratings.

C. Brushless motors and propellers

Most multirotor use brushless outrunner motors because they offer useful torque and power density. Motor speed con-Stant (often expressed as KV in rpm/V) indicates approximate no-load speed per volt; it is not a direct measure of motor quality or lifting ability. Torque, winding resistance, maximum current, thermal limits, rotor size, battery voltage, and propeller load all influence usable thrust. The ESC must be compatible with the motor's electrical and current demands.

Propeller diameter, pitch, blade shape, material, and mass affect static thrust, efficiency, noise, and response. Larger or higher-pitch propellers can increase load and current substantially. Consequently, motors, propellers, ESCs, and battery voltage must be selected as a set using manufacturer thrust/current tables and then verified by a safe static thrust test. A key sizing condition is that available total thrust must exceed vehicle weight with an adequate control margin:

$$\sum_{i=1}^{n} T_i > mg, \tag{1}$$

where T_i is thrust from rotor i, m is takeoff mass, and g is gravitational acceleration. This inequality is necessary but not sufficient; endurance, control authority, thermal limits, and fault tolerance also matter. Propeller damage, an incorrect rotation direction, or an incorrectly assigned motor output can make an otherwise compatible system unsafe.

D. Battery, power module, and distribution

Lithium-polymer (LiPo) and lithium-ion packs are common because of their energy-to-mass ratio. Pack voltage is largely set by cell count; capacity is often stated in ampere-hours (Ah), and approximate stored energy is nominal voltage multiplied by capacity. The pack must support both the ESC/motor current and avionics loads without excessive voltage sag. The C rating is a manufacturer's discharge-current rating, not a substitute for confirming the pack's actual capability, condition, temperature, and wiring.

The power-distribution board (PDB) routes the high-current bus to the ESCs. A power module commonly provides a regulated output for the FC and measures battery voltage/current so the autopilot can estimate consumption and trigger low-battery behavior. Auxiliary regulated rails may power the companion computer, cameras, and range sensors. The high-power propulsion path and low-noise sensor wiring should be routed thoughtfully; poor grounding, voltage transients, electrical noise, and regulator overload can cause sensor errors or unexpected resets. Check connector polarity, wire gauge, continuous current, fuse/protection strategy where applicable, regulator thermal capacity, and brownout behavior before flight.

E. GNSS, GPS, and RTK positioning

GPS is one constellation within the broader GNSS family; receivers may use several constellations, such as GPS, Galileo, GLONASS, and BeiDou. A receiver estimates distance to satellites from their coded radio signals and solves for receiver position and time. Latitude and longitude describe horizontal position in a geographic coordinate reference; altitude depends on the reference surface or height convention used by the receiver. GNSS receivers can also estimate velocity from changes in the navigation solution and Doppler measurements.

Position accuracy depends on satellite geometry, signal blockage, reflections (multipath), atmospheric effects, antenna quality, and receiver design. GPS.gov notes that consumer devices can be within about a 4.9 m radius under open sky, while performance worsens near trees, buildings, and reflective structures; this is an illustrative consumer figure, not a guaranteed drone accuracy specification. Position Hold and waypoint navigation depend on a stable position estimate; RTL also requires a stored home point. A GNSS outage or severe position jump can make these modes inappropriate unless another localization method is available.

Real-Time Kinematic (RTK) positioning uses carrier-phase measurements and correction data from a base station or network to remove common errors. A multi-band receiver such as u-blox ZED-F9P can provide centimeter-class positioning when the correction link, antenna installation, sky view, and

solution status is adequate. RTK improves absolute position but adds integration and operational dependencies: a correction source is required, and a fixed solution can be lost under obstruction or multipath. An RTK receiver does not by itself guarantee centimeter-level mission performance if compass alignment, aircraft attitude, antenna-to-camera extrinsics, or timestamp synchronization are poor.

F. RC transmitter, receiver, and telemetry

A transmitter converts pilot stick and switch movements into a radio control link. A receiver decodes the link and presents channels or serial control data to the FC. Channels commonly represent roll, pitch, yaw, throttle, arm/disarm, and flight mode; additional channels may control a gimbal, payload, or mission option. The receiver and firmware must agree on channel order, failsafe indication, and electrical/serial interface.

ExpressLRS (ELRS) is an example of an RC link ecosystem; it can transport RC control and, with supported configurations, bidirectional MAV Link telemetry on one link. In a conventional setup, RC and GCS telemetry often use separate radio systems. Separate links can simplify functional separation, but antennas and radio frequency compatibility still need attention. RC link loss should trigger the configured RC failsafe, while telemetry loss should trigger the configured GCS failsafe. The system must be tested for the actual armed state and flight mode, not just by checking that the receiver LED is on.

The telemetry system commonly carries MAV Link messages to a GCS such as QGroundControl or Mission Planner. MAV Link is a compact message protocol for commands and vehicle state; telemetry can include position, velocity, attitude, battery status, mission progress, and health information. Telemetry is not merely a video stream, and a high-definition video link is not a replacement for command-and-status telemetry. Bandwidth, latency, antenna placement, interference, range, and encryption/security requirements depend on the mission and region.

G. Companion computer and ROS 2

The companion computer is an onboard general-purpose computer that runs high-level applications. Raspberry Pi-class computers suit modest data-processing and integration tasks, while NVIDIA Jetson-class platforms are designed for edge AI and vision workloads. The Jetson Orin Nano family, for example, targets robotics and computer-vision applications with a configurable power envelope; the required performance must be balanced against mass, heat, and energy consumption. A companion computer may execute image rectification, person detection, visual odometry, point-cloud processing, map construction, obstacle-aware path planning, and autonomous decision logic. ROS 2 provides communication abstractions such as topics for streams, services for request/response operations, and actions for long-running tasks with feedback. This makes it practical to separate nodes for camera drivers, detection, localization, mapping, planner, and vehicle interface.

A common integration path is camera/depth sensor to companion computer over USB or a camera interface; companion software processes the data and exchanges vehicle state and high-level commands with the FC over MAV Link. The FC remains responsible for fast motor control and configured fail safes. The design should explicitly handle startup order, link timeouts, stale image timestamps, CPU saturation, thermal throttling, loss of companion power, and command limits. Autonomy software must avoid sending unbounded or stale targets when the perception pipeline is delayed.

IV. Sensors

A. Inertial measurement unit

An IMU generally combines a three-axis accelerometer and a three-axis gyroscope. The accelerometer measures specific force, including the effect of gravity in a stationary frame; the gyroscope measures angular velocity. The gyroscope responds quickly to rotations but drifts when integrated over time. The accelerometer supplies a gravity reference when non-gravitational acceleration is modest, but vibration and maneuver acceleration can distort its estimate. Sensor fusion balances the complementary strengths of these measurements to estimate attitude and motion.

An IMU is essential to high-rate stabilization because it observes rapid rotational changes more directly than GNSS. It must be calibrated and mounted in a known orientation. Propeller vibration, flexible mounting, temperature drift, clipping at the sensor range, and incorrect axis configuration can corrupt the state estimator. Multiple IMUs can improve fault detection and robustness but do not eliminate the need for calibration and health monitoring.

B. Magnetometer

A magnetometer measures the local magnetic-field vector. Given tilt information, the horizontal component of Earth's field can be used to estimate heading relative to magnetic north; navigation software may apply declination to relate this to true north. Heading errors can result from motor currents, power wires, steel fasteners, speakers, magnets, payload electronics, or current-dependent fields. External compass/GNSS units are often mounted away from the propulsion wiring. Calibration should be performed with the final payload and wiring arrangement, and heading consistency should be monitored during motor operation.

C. Barometer

A barometer measures atmospheric pressure. Since pressure decreases with height, a calibrated pressure-to-altitude relationship provides a useful relative altitude estimate. Weather changes and pressure disturbances around the airframe mean that barometric altitude is not a perfect measurement of height above the ground. Propeller wash, wind, enclosure heating, and blocked pressure ports can cause errors. A barometer is useful for smooth altitude estimation but should be complemented by range sensing for precise low-altitude height above terrain.

D. LiDAR

LiDAR measures distance by emitting laser light and detecting returned light; common ranging designs infer distance from time-of-flight or phase-related measurements. A 2D scanning LiDAR produces ranges in a plane as the scan angle changes. A 3D LiDAR samples multiple vertical and horizontal angles and produces a point cloud. A small 2D unit such as SLAMTEC RPLIDAR A2 illustrates the format of a planar scanner: product variants list roughly 12–16 m measuring ranges, a full 360-degree scan, and UART output, with exact range and sampling depending on variant.

LiDAR is useful for obstacle boundaries, local mapping, localization, terrain/altitude ranging, and geometric scene structure. It generally provides direct range rather than appearance, and can work in low visible light. Limitations include mass and power, moving-part wear on some scanners, sparse returns on small or dark targets, transparent/reflective surfaces, rain/fog/dust, and sunlight effects that vary by wavelength and receiver design. A single plane can miss obstacles above or below the scan plane. For autonomous flight, sensor field of view and blind zones must match the likely obstacle geometry, and raw scans need to be transformed into the aircraft frame with known calibration.

E. Depth camera

Depth describes the distance from the sensor to visible scene points, typically represented as a per-pixel depth image or a 3D point cloud. RGB-D cameras combine color information with depth. Depth can be obtained using stereo correspondence between two viewpoints, structured or projected light, or time-of-flight measurement. Stereo estimates disparity and converts it to range using camera baseline and focal length; error generally increases as range grows or texture and lighting degrade.

The Intel RealSense D455 is one real robotics depth camera. Its published specifications list stereo depth, an RGB sensor, USB 3.1, and a stated operating range of approximately 0.6–6 m, with depth output up to 1280 by 720 at up to 90 frames/. Such a camera can support near-field obstacle detection, object-distance estimation, 3D reconstruction, and visual mapping. Compared with LiDAR it adds texture/color, but its usable range and accuracy depend on exposure, surface texture, sunlight, motion blur, and camera calibration. It should not be assumed to see through dust or fog, and close obstacles may lie inside the minimum valid range.

F. Optical flow

Optical flow estimates apparent image motion between successive frames. A downward-facing camera tracks visual features; together with a downward range measurement, the apparent image displacement can be converted into an estimate of horizontal velocity. PX4 documentation describes this combination as useful when GNSS is unavailable, including indoor and underground environments.

Optical flow can support indoor stabilization, drift reduction, and local velocity estimation using relatively lightweight hardware. It is not an independent global-position solution, and scale typically depends on range information. Low-texture floors, repeating patterns, poor lighting, direct glare, motion blur, large image rotation, and high altitude reduce performance. The estimator should monitor quality metrics and transition safely when the flow output becomes unreliable.

G. Additional technology A: thermal imaging

A long-wave infrared (LWIR) camera senses infrared radiation emitted by objects; an uncooled microbolometer converts small temperature-dependent detector changes into an image. Unlike visible RGB imaging, thermal contrast can reveal a warm person against a cooler background in darkness or partial concealment. Teledyne FLIR's Boson family is an example of a compact LWIR camera module intended for integration in applications including unmanned aircraft.

For search and rescue, thermal frames can provide candidate-person regions at dusk, night, or under weak visible illumination. Advantages include passive sensing and useful contrast when ordinary images are dark. Limitations include lower spatial detail than many RGB cameras, thermal crossover when people and background have similar apparent temperatures, reduced contrast through some materials, weather attenuation, and false positives from animals, warm machinery, or sun-heated surfaces. Thermal detections should be fused with RGB, geometry, or repeated observations and presented with confidence rather than treated as certain human identification.

H. Additional technology B: millimeter-wave radar

Millimeter-wave radar commonly transmits frequency-modulated continuous-wave (FMCW) signals and analyzes the received beat frequency and phase to estimate target range, radial velocity, and sometimes angle. TI's AWR1843 is an example of an integrated 77–79 GHz FMCW radar sensor family developed for range/angle/velocity sensing. Radar can complement cameras and LiDAR where lighting is poor and may provide useful motion cues in light environmental obscurants.

For a drone, radar may detect large obstacles or moving targets and provide a range-rate measurement that is useful for collision-risk estimation. Its advantages include operation without visible light and direct velocity-related measurements. Limitations include clutter, multipath, limited angular resolution in compact low-cost devices, difficulty classifying a person from a small radar signature alone, antenna orientation constraints, and processing complexity. Radar is best treated as complementary sensing, not a complete replacement for geometric depth or visual classification.

V. Communication Interfaces

The physical interface determines how data or control signals are electrically transferred; a higher-level protocol defines the meaning and format of the information. A UART, for example, may carry GNSS messages, receiver control data, telemetry, or LiDAR scans, but those devices do not necessarily share a message protocol. The intended pin assignment, voltage levels,

TABLE II
COMMON INTERFACES IN DRONE AND ROBOTICS SYSTEMS

Interface	Typical use	Example in a drone
UART	Asynchronous serial link for device messages; usually point-to-point.	GNSS receiver, telemetry modem, serial RC receiver, or 2D LiDAR.
I^2C	Short-range two-wire bus for lower-rate sensors and configuration/status data.	Barometer, magnetometer, or a compatible digital power module.
SPI	Synchronous serial bus suited to high-rate local peripheral data.	On-board IMU, flash storage, or sensor module on a custom board.
CAN	Differential multi-node bus designed for robust communication in electrically noisy systems.	Drone CAN GNSS, power modules, or compatible ESCs.
PWM	Pulse-width-coded control output; a signal format rather than a general data bus.	Legacy ESC command input or servo control.
USB	Higher-bandwidth serial data/power interface with host-device roles.	RGB-D camera, companion computer connection, configuration, or storage.

baud rate, bus address, and driver support must be checked for the selected hardware.

The port count and bandwidth available on the chosen flight controller are practical constraints. A high-bandwidth camera normally connects to the companion computer rather than a flight-controller sensor port. Flight-controller-to-companion communication often uses a dedicated UART or USB interface carrying MAV Link messages. Clock synchronization, message rates, cable lengths, electrical noise, and connector strain relief should be considered early, not after assembly.

VI. SYSTEM ARCHITECTURE

Fig. 1 presents a conceptual autonomous quadcopter architecture. Power and command paths are shown separately. The battery feeds the power module/PDB; the high-current bus supplies the ESC bank, while regulated rails supply the FC and companion/perception electronics. Battery voltage/current measurements return to the FC so that energy use can be estimated. The FC accepts GNSS and RC data, exchanges mission/state messages with the GCS through a telemetry radio, and exchanges status and high-level setpoints with the companion computer over MAV Link. Camera and depth/range data terminate at the companion computer, where perception and planning run. The FC then converts valid control demands into four motor outputs.

The principal integration rule is to keep the fast stabilization loop independent from high-level processing. A companion-computer crash should not prevent the FC from responding to its RC failsafe or maintaining a safe state. Similarly, obstacle avoidance needs an explicit path from perception output to a control setpoint and a defined timeout when fresh sensor data is unavailable. A camera or LiDAR without calibrated transforms, timestamps, and a tested software consumer is not an operational obstacle-avoidance system.

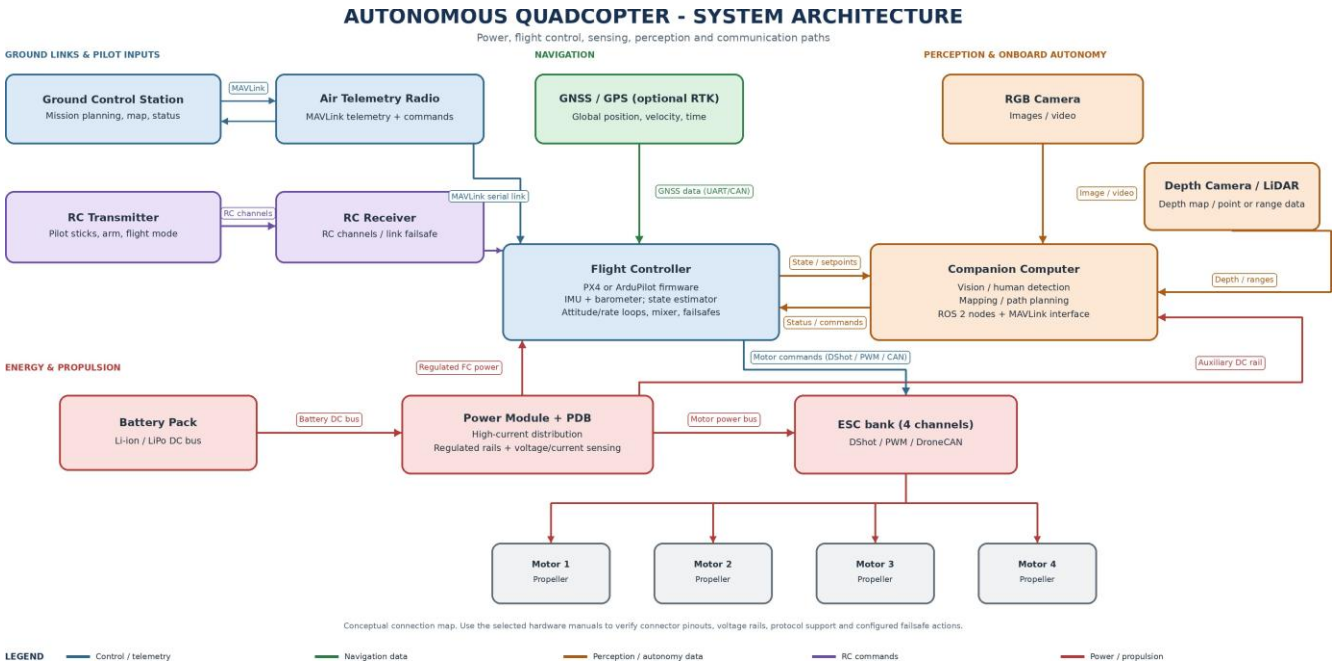
The architecture should be verified in stages. First, test power rails and sensor communications with propellers removed. Second, verify sensor orientation/calibration, RC channel mapping, and motor order/direction with ESC outputs disabled or a safe fixture. Third, test actuator response and vibration in a controlled environment. Then evaluate position hold, waypoint following, telemetry loss, RC loss, low-battery response, GNSS degradation, companion-computer restart, and perception dropouts in simulation and a controlled flight-test plan. Logs should be reviewed after every stage. This sequence is important because many integration faults appear as plausible but incorrect sensor data rather than obvious electrical failures.

VII. COMPONENT SELECTION FOR A SEARCH-AND-RESCUE MISSION

The mission requires outdoor flight, waypoint navigation, obstacle detection, person localization, telemetry to a GCS, and onboard image processing. A balanced reference design is shown in Table III. The choice favors compatible and maintainable parts rather than maximum compute or sensor range. The listed models are examples; a final bill of materials should be matched to takeoff mass, required endurance, weather envelope, field servicing, and regulatory limits.

A suitable operational concept is to plan broad search waypoints in the GCS, fly them using GNSS navigation, and use the companion computer to process imagery continuously. When a candidate person is detected, the computer should combine image coordinates with vehicle pose, camera calibration, and either known terrain or range/depth data to estimate a location; it should send the candidate location, confidence, and timestamp to the GCS. The system should preserve operator oversight and provide a means to pause or cancel autonomous behavior. Obstacle avoidance needs conservative speed limits, a defined stopping/turning margin, and a recovery behavior if sensor data becomes stale.

The D455-class depth camera is adequate for near-field prototyping, but its limited stated range makes it inappropriate as the sole sensing input at high speed or in poor visibility. A 2D scanning LiDAR may be considered for planar geometry, while a 3D LiDAR or radar is a stronger but typically more costly complement where the mission and payload permit. Thermal imaging adds useful low-light cues but can confuse warm rocks, animals, and equipment with people; RGB and repeated observations reduce, but do not eliminate, false detections. RTK should be treated as an accuracy improvement rather than a guarantee: coverage, corrections, antenna setup, and estimator status remain part of the mission design.



VIII. Integration Risks and Verification Priorities

The dominant risks arise at interfaces. Electrical compatibility should be confirmed before connecting battery power: ESC input voltage must match pack voltage, power-module output must meet FC requirements, and companion peripherals must not overload the 5 V or 12 V rail. Current measurements should be checked against a known load, and the battery estimate should be compared with measured consumption. The motor/propeller combination should be validated against thrust and current data rather than selected from motor KV alone.

Software interfaces also need contracts. Document which device owns each serial port and which protocol/rate it uses. Define the coordinate frames and units for position and obstacle data. Ensure that camera time, FC time, and GNSS time are related well enough for localization, and that the camera's pose relative to the aircraft is calibrated. Use watchdogs and heartbeats to detect stale messages. Limit companion-issued setpoints to a tested envelope and reject impossible or expired commands.

Before any outdoor deployment, verify each failsafe and the behavior of combined failures in simulation or a controlled test plan. Test loss of RC, loss of GCS telemetry, loss of GNSS, low battery, estimator faults, companion-computer loss, and depth-sensor dropout. Choose the response based on mission risk: RTL can be suitable with a reliable home estimate and clear return path, while Land or a controlled hold may be safer when navigation quality is poor. The safest response is aircraft- and environment-dependent; it must be configured, demonstrated, and documented, not assumed.

IX. Conclusion

An autonomous quadcopter combines a real-time autopilot, propulsion electronics, energy distribution, positioning, communications, perception sensors, and a companion computer. The flight controller performs deterministic stabilization and owns actuator authority; the companion computer enables computationally intensive perception and planning through a well-defined MAV Link interface. GNSS supports global navigation, inertial and pressure sensors support state estimation, and LiDAR/depth/optical-flow technologies add local geometry or velocity estimates. Thermal imaging and mm Wave radar can complement visible cameras under difficult lighting or provide motion/ranging cues, but each has limitations that must be handled explicitly.

The architecture and search-and-rescue selection demonstrate that the best design is a compatible system with known limitations, realistic power and thrust margins, distinct RC and telemetry roles, clear ownership of control functions, and tested failsafe behavior. Reliable autonomy is an end-to-end property created by correct integration, calibration, monitoring, and verification rather than by any single sensor or processor.

TABLE III
ILLUSTRATIVE SEARCH-AND-RESCUE COMPONENT SELECTION

Requirement	Selected component / technology	Reasoning and trade-offs
Flight controller	Pixhawk 6C-class autopilot running PX4 or ArduPilot	A mature autopilot class with established GNSS, RC, actuator, and telemetry integration. It can retain stabilization and failsafe authority if companion software fails. Confirm exact firmware target, supported ESC protocol, sensor set, and power module compatibility.
Companion computer	NVIDIA Jetson Orin Nano-class edge computer	Provides an onboard platform for RGB/thermal inference, point-cloud processing, and planning. It is more capable than a small microcontroller, but adds power, cooling, storage, boot-time, and software-maintenance requirements [13].
Positioning	Multi-constellation GNSS with optional RTK corrections, e.g., u-blox ZED-F9P class	GNSS supports global waypoints and RTL; RTK can improve precision when correction data and satellite visibility are adequate. A non-RTK mode should still be handled safely, and magnetic interference/antenna placement need validation [8], [9].
Obstacle detection	Forward-facing stereo depth camera such as Intel RealSense D455; optional range/LiDAR redundancy	Supplies dense local depth for near-field obstacle geometry in a compact package. The stated range is approximately 0.6–6 m; bright sunlight, low texture, weather, and limited range mean it should not be the only protection for higher-speed flight [12].
Human detection/localization	RGB camera plus compact LWIR thermal camera	RGB supports appearance-based person detection and localization in daylight; thermal can improve candidate detection at night. Fuse detections with vehicle pose and sensor calibration, show confidence, and avoid representing detection as a guarantee [16].
Communication	ELRS-compatible RC receiver for pilot override plus, a separate MAV Link telemetry radio to the GCS	Separates low-latency pilot input from mission/status telemetry and makes failures easier to reason about. A combined link is also possible, but link-loss behavior and data priority must be validated [10], [17].

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